SAIKIRAN CHEPA

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Professional Summary

Embedded DSP and AI/ML Engineer with 10+ years of experience delivering measurable performance gains across the full engineering stack—from bare-metal assembly on ARM, TI DSP, and Qualcomm Hexagon platforms to training and deploying deep learning models on both edge devices and cloud infrastructure. Adept at bridging low-level systems engineering (fixed-point arithmetic, SIMD intrinsics, Linux system programming) with modern AI practices (LLMs, RAG, RLHF), with a track record of reducing compute cycles by up to 60% and shipping production-ready AI systems end-to-end. A proactive, self-directed learner evidenced by multiple advanced certifications and a GATE DA 2026 qualification; recognized for clear technical communication, collaborative problem-solving across cross-functional teams, and the ability to adapt rapidly to new architectures, toolchains, and project requirements.

Key Highlights

- Reduced DSP compute cycles by ~60% through float-based loop optimizations in a commercial CarVoice pipeline.
- Closed ARMv8-A vs. ARMv7-A performance gap from 2.2× to 1.4× using 25 hand-optimized ARM NEON functions.
- Trained ResNet18 on CIFAR-10 (~90% top-1 accuracy) and deployed it on Qualcomm Hexagon DSP, GPU, and CPU.
- Designed and shipped a production-ready Intelligent Email Query Assistant on AWS using Gemini LLM, WebSockets, and MCP.
- GATE DA (Data Science & AI) 2026 Qualified — demonstrating rigorous theoretical grounding in ML, probability, and statistics.
- NPTEL Domain Scholar in Signal Processing & Communications — 60-week IIT-level program across 7 advanced courses.
- Deep expertise in Linux system programming: POSIX threads, IPC mechanisms, memory/process management, and device drivers.

Professional Experience

Embedded AI Engineer *Zentree Labs, Hyderabad* **Jan 2026 – Present**

- Trained ResNet18 CNN on CIFAR-10 using PyTorch Lightning; achieved ~90% top-1 accuracy with systematic hyperparameter tuning.
- Built full deployment pipeline: PyTorch → ONNX → QNN compilation (`qnn-onnx-converter`) → shared library generation (`qnn-model-lib-generator`).
- Deployed model on Snapdragon 710 across CPU, GPU, and Hexagon DSP; automated backend evaluation.
- Cross-compiled ARM64 host application with Android NDK.
- Resolved NumPy/QAIRT compatibility issues and NDK toolchain configuration challenges.
- Quantization of LLMs using Bitsandbytes(Aware of AWQ, SmoothQuant, QAT)

DSP Engineer — *Capgemini (Client: Qualcomm), Hyderabad* **Dec 2016 – May 2023 | Jan 2025 – Mar 2025**

- Converted floating-point speech algorithms to fixed-point; implemented `log10` and `pow` using fixed-point primitives `log2`/`pow2`.
- Used MATLAB Codegen to produce C code; refactored MATLAB scripts to resolve codegen errors and replaced dynamic arrays with fixed-size implementations for embedded constraints.
- Developed CSIM APIs for MATLAB-based DSP models, enabling software-in-the-loop validation.
- Optimized stereo FIR filter with decimation-by-2 in assembly by skipping alternate sample computation and computing both channels simultaneously in the innermost loop to maximize instruction slot utilization.
- Implemented upsampling followed by FIR filtering.
- Automated build, compile, and test pipelines using Python scripts, batch/shell scripts, and CMake.
- Optimized matrix multiplication for AGVC (AI/ML-based speech codec) targeting Intel AVX2 architecture.

DSP Engineer — *Capgemini (Client: Goodix), Bangalore* **Aug 2023 – Jul 2024**

- Integrated RISC-V DSP library FFT/IFFT into the VoiceExperience codebase; validated non-bitexact outputs using MATLAB RMSE analysis (in dB) and Adobe Audition waveform inspection.
- Bridged SHARC floating-point FFT/IFFT library into CarVoice's fixed-point pipeline via float↔fixed conversions, enabling reuse of mature library routines.
- Optimized performance-critical inner loops by processing in floating-point and converting back to fixed-point, achieving ~60% cycle reduction.
- Closed ARMv8-A vs. ARMv7-A performance gap from 2.2× to 1.4× by coding 25 hot functions with ARM NEON intrinsics.
- Developed Python scripts to compare ARMv7-A and ARMv8-A flat profiles and automatically identify functions not yet optimized for ARMv8-A.
- Measured MCPS on Android devices using ADB; profiled at function level using Simpleperf.
- Automated MCPS benchmarking on Raspberry Pi 4: extracted minimum of 5 runs per frame, then derived average and peak MCPS across the full dataset.

Senior DSP Engineer *Couth Infotech* **Apr 2012 – Mar 2015**

- Optimized SILK, Speex, and Opus codecs for TI C66x and C64x+ DSPs using linear assembly.
- Partitioned scratch and channel buffers into 4 KB blocks; reduced stack usage through scratch-based memory management.
- Set up Network Development Kits (NDK) for C6678 and C6472 boards; validated codec against standard and non-standard test vectors.
- Authored compliance wrappers for: DataMove (runtime-relocatable channels and tables), illegal read/write detection, register preservation, buffer corruption detection, I/O alignment, interrupt testing, stack and buffer sizing, code coverage, and scratch contamination checks.

Senior Software Engineer *Aricent* **Jul 2005 – Jul 2009**

- Optimized eAAC+ decoder for TI C64x+ DSP; developed both interleaved and non-interleaved output modes.
- Ported iLBC codec from 32-bit floating-point to fixed-point representation.
- Optimized audio and speech codecs across TI C64x+ and C66x platforms.

Linux System Programming

Hands-on experience with core Linux and POSIX programming concepts essential for embedded, systems, and driver development:

- **Multithreading & Synchronization:** POSIX threads (pthreads), thread lifecycle management, mutexes, condition variables, semaphores, and deadlock avoidance strategies.
- **Memory & Process Management:** Virtual memory, `mmap`, dynamic memory allocation, process creation (`fork/exec`), zombie/orphan handling, and signals.
- **IPC Mechanisms:** Named pipes (FIFOs), POSIX shared memory, message queues, and POSIX/BSD sockets for inter-process communication.
- **Device Drivers:** Character device driver development, kernel module programming (`insmod/rmmod`), `/proc` and `/sys` filesystem interfaces.
- **Build Infrastructure:** GNU toolchain (gcc, ld, objdump, nm, readelf), Makefile authoring, cross-compilation toolchains, and building the Linux kernel from source (menuconfig, modules, device tree overlays).

Education

- **MTech in Electrical Engineering** *IIT Madras* 2005
- **BTech in Electronics and Communication Engineering** *GRIET, JNTU Hyderabad* 2003

Certifications & Achievements

GATE DA (Data Science & AI) 2026 | *Qualified*

Domain Scholar – Signal Processing and Communications (NPTEL, 60 Weeks)
- Discrete Time Signal Processing, Multirate DSP, Adaptive Signal Processing
- Applied Linear Algebra, Probability Foundations for Electrical Engineers, Introduction to Information Theory, Principles of Signals and Systems

Mathematics for Machine Learning and Data Science Specialization — Coursera
- Linear Algebra, Calculus, and Probability & Statistics for ML
- Key skills: Bayesian Statistics, Dimensionality Reduction, Sampling, Numerical Analysis, NumPy

Deep Learning Specialization — Coursera
- Designed and trained CNNs, RNNs, and transformer-based models using TensorFlow
- Applied hyperparameter tuning, bias-variance analysis

ERA V3 (Extensive Reimagined AI, v3) — The School of AI, Bangalore
- PyTorch hands-on: neural networks, CNNs, ResNets, and optimization techniques
- Trained transformer models from scratch; studied LLM fine-tuning strategies: SFT, RLHF, PPO, DPO, GRPO
- Multi-modal AI, AI agents, Model Context Protocol (MCP), and Retrieval-Augmented Generation (RAG)
- **Capstone Project: Intelligent Email Query Assistant**
  - Built a client-server application enabling natural-language querying of personal Gmail accounts
  - Client-side Gmail API OAuth authentication; Gemini LLM backend
  - Applied regex-based filtering of promotional emails; limited results to top 20 most relevant messages
  - Implemented server-side transaction extraction pipeline and MCP-based external calculator orchestration

Technical Skills

- **Programming Languages:** C, Python, MATLAB, Assembly (ARM NEON, TI Linear Assembly, QDSP intrinsics)
- **Embedded & DSP:** Fixed-point arithmetic, FIR/IIR filters, FFT/IFFT, multirate DSP, floating-to-fixed conversion, CSIM API development, codec optimization (SILK, Speex, Opus, eAAC+, iLBC, AGVC)
- **SIMD & Performance Optimization:** ARM NEON intrinsics, Intel AVX2, MCPS profiling, cycle reduction, Simpleperf, ADB-based benchmarking
- **Processor Architectures:** ARMv7-A, ARMv8-A/ARM64, Qualcomm Hexagon DSP, Intel AVX2, RISC-V DSP, TI C64x+, TI C66x, SHARC DSP
- **Linux System Programming:** POSIX threads, mutexes, semaphores, shared memory, named pipes, sockets, device drivers, kernel modules, Makefile, GNU toolchain, kernel build from source
- **AI & Deep Learning:** CNNs (ResNet18), RNNs, Transformers, LLMs, RAG, multi-modal AI, edge AI deployment, fine-tuning (SFT, RLHF, PPO, DPO, GRPO)
- **Frameworks & Libraries:** PyTorch, PyTorch Lightning, ONNX, Qualcomm QNN SDK, Qualcomm SNPE, MATLAB Codegen, Hugging Face Transformers
- **Cloud & Deployment:** AWS EC2, AWS SageMaker, AWS S3, cross-compilation (ARM64, Android NDK)
- **Build & Automation:** CMake, Makefile, Git, SVN, Python scripting, shell/batch scripting, adb
- **Tools & Environments:** VS Code, Cursor AI, Claude Code, CCS, Eclipse, DS-5, LangGraph, MCP, OpenAI SDK, Windows, Linux, Android, Raspberry Pi